How much can a causal-discovery benchmark resolve? Sampling noise, reporting choices and the simulation gap on real data

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Abstract

Recent methods for causal discovery routinely report gains of a few accuracy points on the Tübingen cause–effect pairs or a few edges on the Sachs protein network. We ask what these benchmarks can resolve. On the 102 univariate Tübingen pairs, twelve direction rules have accuracies from 40% to 65%, and only one of 66 paired comparisons survives a Holm correction; a paired test at $n = 102$ has 80% power only for differences of about 19 points. A multiverse of 32 defensible reporting choices (official weights or none, subsets by sample size, evaluation on the most confident 25 to 100% of pairs) moves the accuracy of a single method by up to 47 points, and seven of twelve methods are ranked first under at least one choice. Trying ten random subsets of 50 pairs raises the expected best accuracy by seven to eight points. On the Sachs network, 16 of 22 methods have a bootstrap interval that contains the best F_1, and the smallest significant difference is 0.17. Methods trained on simulated data lose the most: a pair classifier falls from 95% on its simulator to 40% on Tübingen. We trace the apparent progress to these four sources, none of which involves a fault of the original authors, and give a reporting checklist: paired tests with a power statement, the full coverage curve, subsets fixed before the data are seen, and a statement of the effective sample size. We re-implemented four methods from 2025–2026 preprints and ran nine published packages at their defaults; none is distinguishable from the best simple baseline.

1 Introduction

New causal-discovery methods are usually accompanied by a table in which the new method is best. On pairs the table is the Tübingen collection of 108 cause–effect pairs [Mooij et al., 2016], 102 of which are univariate; on graphs it is the Sachs protein network [Sachs et al., 2005] with 17 consensus arcs, or simulated graphs. We do not question any particular table. We ask a prior question: with the data these benchmarks contain, which differences between methods can be detected at all, and how far can a correct but unconstrained choice of protocol move a reported number?

The question is not academic. A recent same-hands re-evaluation of bivariate rules on Tübingen with a parameter-free compression baseline reports numbers close to ours for RECI and IGCI [Stienstra, 2026], and the gap between published and re-measured accuracies is a recurring observation in the literature on simulated graphs [Reisach et al., 2021]. We add three things: a quantitative resolution analysis on real data (Section 3), a multiverse over reporting choices that shows how large the room for selection is (Sections 4, 5), and evidence that learned methods lose their advantage when the test data differ from the simulator (Section 6). The methods compared were all run by us under one protocol; they are listed in Section 2.

What this paper does not claim. We did not audit any published table, we do not know which protocol an author used, and a method that is not distinguishable from a baseline here may still be better on other data. The multiverse is a set of choices that a reader of a paper could face; it is not a statement about what authors did.

2 Data and methods

Pairs. The 102 univariate Tübingen pairs with the official weights (the weights downweight groups of similar pairs). Preprocessing is fixed: at most 2000 random rows, each variable dequantised with a small uniform jitter.

Methods. Twelve direction rules, all run by us at settings fixed before the evaluation: IGCI [Daniušis et al., 2010], RECI [Blöbaum et al., 2018], an additive noise model with a spline regression and an HSIC test, the pairwise LiNGAM statistic [Hyvärinen and Smith, 2013], a Gaussian-process additive noise model, LOCI [Immer et al., 2023] with the authors' code, a classifier trained on simulated pairs (PDM, ours), sorted-conditional compression [Stienstra, 2026] on dequantised and on raw rows, topological residual asymmetry [El Bouchattaoui, 2026], AVICI [Lorch et al., 2022] with its pretrained model on two-variable input, and RESIT with a random forest. Compression, TRA and AVICI/RESIT are recent; compression and TRA were implemented from the preprints because no code was available (TRA's constants are our choice).

Graphs. The Sachs observational data (853 cells, 11 proteins, 17-arc consensus network) and the Causal Chambers [Gamella et al., 2025] wind- and light-tunnel recordings with a known graph. Twenty-two graph methods are on Sachs (PC, GES, DirectLiNGAM, NOTEARS, DAGMA, PCMCI+, GOLEM, NOTEARS-MLP, GraN-DAG, DAG-GNN, GRaSP, BOSS, RESIT, CAM-UV, AVICI, BF-BIC with BOSS [Ramsey et al., 2025], and our own PC-GCM, BTAD, LRG and sequential certificates). The full protocol and all per-method tables are in the companion evaluation (`paper/bench` in the repository).

Pre-registration. Each evaluation script was committed before it was run (experiments 42–48 in the repository); the analysis in this paper uses the stored per-pair decisions of those runs.

3 Sampling noise

Table 1 gives accuracies with exact Clopper–Pearson intervals; the half-width is about 0.10 for every method. Two methods that agree on 59% of the pairs can differ in accuracy, and the exact McNemar test on the discordant pairs is the correct comparison because the methods are scored on the same pairs. Of 66 pairwise comparisons, 11 have a raw $p < 0.05$ and one survives Holm ($p = 0.041$). Simulating a paired comparison with the observed discordance rate (41%), a difference of 0.19 in accuracy is needed for 80% power. A reported gain of two to five points on Tübingen is therefore far below what the collection can resolve, whatever the method.

4 A multiverse of reporting choices

We evaluate every method in the 32 combinations of (i) official weights or none, (ii) all pairs or only pairs with at least 250, 500 or 1000 rows, and (iii) accuracy on all pairs or on the 75, 50 or

Table 1: Twelve pair rules on the 102 Tübingen pairs (unweighted accuracy, 95% interval). Multiverse: the range of accuracy over 32 reporting universes, the best and worst rank, the number of universes in which the method ranks first (ties share the rank), the number in which it is above RECI, and the expected best accuracy after trying ten random subsets of 50 pairs.

Method	Accuracy [interval]	Multiverse range	Rank	First	>RECI	Best of 10 subsets
IGCI	0.559 [0.46, 0.66]	0.27–0.75	3–12	0	0	0.637
RECI	0.608 [0.51, 0.70]	0.58–1.00	1–9	14	0	0.681
ANM-HSIC	0.608 [0.51, 0.70]	0.53–0.91	1–12	3	8	0.684
Pairwise LiNGAM	0.608 [0.51, 0.70]	0.52–0.90	3–12	0	1	0.682
PDM	0.402 [0.31, 0.50]	0.33–0.71	8–12	0	0	0.478
LOCI	0.647 [0.55, 0.74]	0.51–0.91	1–12	3	6	0.720
ANM, Gaussian process	0.554 [0.45, 0.65]	0.55–0.95	1–11	6	9	0.629
Compression, dequantised	0.608 [0.51, 0.70]	0.61–1.00	1–10	8	9	0.682
Compression, raw rows	0.647 [0.55, 0.74]	0.58–0.96	1–10	1	10	0.721
TRA	0.618 [0.52, 0.71]	0.56–0.97	2–10	0	2	0.685
AVICI	0.588 [0.49, 0.68]	0.41–0.83	1–12	3	7	0.665
RESIT	0.613 [0.52, 0.71]	0.61–0.75	2–10	0	7	0.686

25% most confident decisions (by the absolute score of the method). Each is a choice that has been made in the literature on pairs for a reason: weights are the benchmark's recommendation, a row threshold avoids tiny pairs, and decided-only accuracy reflects a method that abstains. The same method then spans a wide range of accuracy (Table 1): 0.27 to 0.75 for IGCI, 0.51 to 0.91 for LOCI, 0.55 to 0.95 for the Gaussian-process ANM, 0.58 to 1.00 for RECI. Seven of the twelve methods are first in at least one universe; RECI is first in 14 of 32, compression in 8, the Gaussian-process ANM in 6, and four more in 3 or 1. Ranks of a single method run over most of the list, for example from first to twelfth for the LOCI. In one universe, any of seven methods can be called the best; the evidence for a ranking is a property of the protocol as much as of the methods.

Two single choices are large in themselves. The official weights move RECI from 60.8% to 70.9% and compression from 60.8% to 68.0%, while for LOCI they move it down (64.7% $\rightarrow$ 63.9%). Raw against dequantised rows moves compression from 64.7% to 60.8% unweighted.

5 Selective subsets

A common practice in pair benchmarks is to evaluate a method on a subset of pairs (for example those with a given variable type or sample size). We drew 10000 random subsets of 50 pairs. The accuracy of a method on a subset has a standard deviation of about 0.05 and for the seven methods whose full-sample accuracy is within 0.04 of RECI the probability of beating RECI on a subset lies between 0.43 and 0.68. The expected best accuracy over m independently drawn subsets, which is what is reported by a researcher who tries m subsets and keeps the best, grows with m: for RECI 0.608 ($m = 1$), 0.648 (3), 0.681 (10); for LOCI 0.647, 0.688, 0.720. The inflation at $m = 10$ is seven to eight points, and the same for every method, so a ranking is not changed by it; what changes is the claim of an absolute level.

6 The simulation gap

Methods trained on simulations lose the most. Our own pair classifier (PDM) has 95% accuracy on its simulated test and 83% on a different simulated family, and 40.2% on Tübingen ($p = 0.059$

against chance; 0.93 and 0.84 on the two chamber pair sets). Our graph classifier (LRG) has $F_1 = 0.86$ on simulated graphs and 0.36–0.53 on the chambers. The pretrained AVICI model reaches 90% on pairs from independent-row chamber recordings but 58.8% on Tübingen and 55.7% on pairs from random-walk recordings. We cannot say which feature of the real data is responsible; the simulators contain serial dependence, drift and several mechanism families and still do not cover them.

7 Graph benchmarks

On Sachs the same analysis gives the same picture. For 22 methods, 16 have a 95% bootstrap interval (over the 55 variable pairs) that contains the best F_1 (0.67) and 7 have a point estimate within 0.10 of it. Of 231 pairs of methods, 62 differ significantly and the smallest significant difference in F_1 is 0.17. The reference has 17 arcs, so one edge changes recall by 0.06. In the Causal Chambers, the bipartite graph (actuators to sensors) makes the skeleton mostly feature selection, and in 17 of 19 new wind-tunnel recordings the actuators are random walks with an effective sample size of about three out of 4000 rows, where no method has strict F_1 above 0.42. Several neural methods time out at sixteen variables. Appendix tables of the companion evaluation give the per-method values.

8 Why published gains can look larger

Combining the evidence, a published gain on a small real benchmark can arise without any change in the method's true accuracy from (i) the resolution limit: two methods that differ by two to five points are statistically the same on 102 pairs; (ii) free reporting choices that move one method by up to 47 points; (iii) selection among subsets or settings, worth seven to eight points after ten attempts; and (iv) the gap between a simulator and real data for learned methods. None of these requires an error by an author; each is a degree of freedom the benchmark does not constrain. These are mechanisms, not an accounting of a particular paper.

9 Recommendations

(1) Compare methods with a paired exact test on the same pairs, state the power, and do not interpret a difference below the detectable one. (2) Report the full coverage curve, not one decided-only number, and report weighted and unweighted accuracy. (3) Fix subsets, preprocessing and settings before the data are seen, and publish the code that does it. (4) State the effective sample size of every real data set, and do not use a benchmark whose effective size is a few observations to rank methods. (5) For learned methods, report accuracy on real data next to the simulated test, with the simulator's overlap with the real data described. (6) Prefer benchmarks with more independent units: interventional recordings and collections with many more pairs would lower the detectable difference.

10 Limitations

The study uses 102 pairs and one protein network, which is itself the resolution limit it describes. The twelve methods were run by us; some at library defaults, and tuning may change a ranking. The multiverse is finite and its choices are ours. We did not run CAM, SCORE or PNL (PNL needs

880 s per pair), and two preprints were implemented from their description. The simulation-gap evidence uses our own learned models and one pretrained model.

Data and code. All code, per-pair decisions and scripts are in the repository `vinhqdang/SPRINT-CD`. The Tübingen pairs, the Sachs data and the Causal Chambers data are public.

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